Insurance Charges

Data Analysis & Preprocessing Report

Medical Cost Personal Dataset · Kaggle

Objective: Preprocess data for insurance charges regression modelling

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1. Dataset Overview

The dataset used is the Medical Cost Personal Dataset, a widely used benchmark dataset for regression problems in healthcare analytics. It contains 1,338 records of individual insurance policyholders across the United States, with 7 columns covering demographic, lifestyle, and financial information.

1.1 Column Reference

Column	Type	Description
age	Numeric (int)	Age of the primary beneficiary (18–64)
sex	Categorical	Gender of the policyholder — male / female
bmi	Numeric (float)	Body Mass Index — ratio of weight to height squared
children	Numeric (int)	Number of dependents covered by insurance (0–5)
smoker	Categorical	Whether the beneficiary smokes — yes / no
region	Categorical	Residential region in the US — southwest, southeast, northwest, northeast
charges ★	Numeric (float)	Individual medical costs billed by the insurer — TARGET VARIABLE

1.2 Raw Dataset Statistics

Total Rows	Total Columns	Numeric Columns	Categorical Columns
1,338	7	4 (age, bmi, children, charges)	3 (sex, smoker, region)

No missing values were found in any column. One exact duplicate row was identified and removed, leaving 1,337 clean records for analysis.

2. Exploratory Data Analysis (EDA)

EDA was performed to understand the shape of the data, identify relationships between variables, and form hypotheses before any transformation was applied.

2.1 Numeric Feature Distributions

Histogram plots with KDE curves were generated for all four numeric columns. Key observations:

- age: Roughly uniform distribution between 18 and 64. No extreme skew.
- bmi: Near-normal distribution centred around 30. A small number of high-BMI outliers exist above the upper IQR fence.
- children: Discrete count variable (0–5). Majority of policyholders have 0–2 children. A countplot was used instead of a histogram for better readability.
- charges: Heavily right-skewed (skewness ≈ 1.5). Most policyholders pay under $15,000, but a long tail extends to $64,000. This requires treatment before regression modelling.

2.2 Categorical Feature Distributions

Countplots were generated for sex, smoker, and children:

- sex: Near-equal split — 676 males and 662 females. No class imbalance.
- smoker: Imbalanced — 1,064 non-smokers (79.5%) vs 274 smokers (20.5%). Smoking is a minority class but, as confirmed later, the single strongest predictor of charges.
- region: Roughly equal distribution across all four US regions (~325 each).

2.3 Smoker vs Charges — Key Finding

A box plot of smoker vs charges revealed the most important insight in the dataset:

- The minimum charge for a smoker (~$13,000) is higher than the 75th percentile charge for non-smokers.
- Non-smoker charges are low and predictable (narrow IQR). Smoker charges are high and highly variable (wide IQR), suggesting that smoking interacts with other factors.
- Non-smokers with charges above the whisker are likely driven by age, high BMI, or other comorbidities.

This visual finding was later confirmed statistically by the Pearson correlation test ($r = 0.787$, $p < 0.001$).

2.4 Correlation Heatmap

A Pearson correlation heatmap was generated on all numeric columns. Key takeaways:

- charges and age: Moderate positive correlation ($r \approx 0.30$).
- charges and bmi: Weak positive correlation ($r \approx 0.20$).
- charges and children: Near-zero correlation — children is largely uninformative on its own.

- No negative correlations found. No strong feature-to-feature correlations, suggesting low multicollinearity in the raw numeric features.

3. Data Cleaning

Data cleaning was performed on a copy of the raw dataset (data_cleaned = data.copy()) to preserve the original data for reference. Three cleaning operations were applied:

3.1 Missing Value Check

A null value check was run across all 7 columns using isnull().sum(). Result: zero missing values in any column. No imputation was necessary.

3.2 Duplicate Removal

An exact row-level duplicate check was performed using drop_duplicates(). One duplicate row was identified and removed, reducing the dataset from 1,338 to 1,337 rows. Retaining duplicates would double-count that patient and bias the model toward their specific profile.

3.3 Categorical Label Verification

A value_counts() check was performed on all categorical columns (sex, smoker, region) to verify that no inconsistent labels existed (e.g., 'Male' vs 'male' vs 'M'). All values were clean and consistent — no standardisation was required.

3.4 Cleaning Summary

Check	Finding	Action
Missing values	None found	No action needed
Duplicate rows	1 duplicate found	Removed with drop_duplicates()
Label consistency	All labels clean	No standardisation needed

4. Outlier Detection & Treatment

Outliers were detected visually in the EDA boxplot phase. The IQR capping method was applied to treat them after cleaning.

4.1 Method — IQR Capping

The standard IQR fencing formula was used to compute the acceptable range for each column:

Lower fence = Q1 − 1.5 × IQR | Upper fence = Q3 + 1.5 × IQR

Values below the lower fence were replaced with the lower fence value. Values above the upper fence were replaced with the upper fence value. This is called clipping — the row is preserved, but the extreme value is capped at the boundary.

Capping was chosen over row deletion deliberately. Deleting outliers discards real patient data. Capping acknowledges that the value is extreme while preventing it from dominating the model's loss function.

4.2 Columns Treated

Only bmi and charges were found to have meaningful outliers. Age and children had clean distributions within natural bounds.

Column	Lower Fence	Upper Fence	Original Max	After Capping
bmi	Q1 − 1.5×IQR	Q3 + 1.5×IQR	53.13	~47.4 (capped)
charges	Q1 − 1.5×IQR	Q3 + 1.5×IQR	$63,770	~$51,194 (capped)

A before/after boxplot was generated to visually confirm that the outlier dots above the upper whisker were removed after capping.

Importantly, capping was applied to charges before the log transform — because if extreme values remain in charges when log() is applied, they persist as extremes in log-space (less severe, but still present).

5. Feature Engineering & Encoding

Machine learning models are mathematical functions — they cannot process text strings. All categorical columns were converted to numeric representations. One new feature was created using domain knowledge.

5.1 Label Encoding — Binary Columns

Label encoding converts a binary text column directly to 0 and 1. This is appropriate only when there are exactly two categories and no implied ordering.

- sex → is_female: male mapped to 0, female mapped to 1. Column renamed to is_female for semantic clarity.
- smoker → is_smoker: yes mapped to 1, no mapped to 0. Column renamed to is_smoker.

Both columns were verified via value_counts() before encoding to confirm exactly two unique values existed with no typos or inconsistencies.

5.2 One-Hot Encoding — Region

The region column has four categories: southwest, southeast, northwest, northeast. Label encoding was not used here because it would impose a false numerical ordering (e.g., telling the model region 3 > region 1). Regions have no inherent hierarchy.

pd.get_dummies() was applied with drop_first=True to avoid the Dummy Variable Trap. Keeping all four encoded columns would introduce perfect multicollinearity — the model can always infer the dropped category when all others are 0. This produced three new columns: region_northwest, region_southeast, region_southwest (region_northeast is the reference).

5.3 BMI Categorisation — Feature Engineering

BMI was transformed from a continuous numerical variable into official WHO medical categories using pd.cut() with clinically established thresholds:

Category	BMI Range	Encoded Column	drop_first
Underweight	< 18.5	Dropped (reference)	Yes
Normal	18.5 – 24.9	bmi_category_Normal	Kept
Overweight	25.0 – 29.9	bmi_category_Overweight	Kept
Obese	≥ 30.0	bmi_category_Obese	Kept

This allows the model to detect threshold effects — for instance, that crossing from Overweight to Obese carries a distinct charge penalty — which a raw continuous BMI value cannot express as clearly.

6. Target Variable Analysis

6.1 Skewness Problem

The raw charges column has a skewness of approximately 1.5 — indicating heavy right skew. Most policyholders pay between $1,000 and $15,000, but a long tail extends to $63,770 (pre-capping).

This poses a problem for regression models. The least squares loss function squares each prediction error. A single extreme charge value produces a squared error that can dominate the entire loss calculation, causing the model to bend towards outliers and perform poorly on the majority of the data.

6.2 Log Transform

A natural log transform was applied to charges to correct the skewness:

log_charges = ln(charges)

The log function compresses large values proportionally more than small ones, pulling the long tail inward. After transformation, skewness drops from ~1.5 to approximately 0.2 — a near-symmetric distribution.

The original charges column was preserved in data_cleaned. The new log_charges column is used as the modelling target. When interpreting model predictions, the inverse transform (np.exp()) is applied to convert back to actual dollar values.

6.3 Why This Is Not Data Leakage

A log transform on the target is a scale transformation, not an information injection. It does not use target values to create or influence features — it only changes the unit of measurement the model works in. The underlying patient data is unchanged. This is standard practice for skewed regression targets such as house prices, salaries, and insurance costs.

7. Statistical Feature Selection

Two complementary statistical tests were used to formally evaluate each feature's relationship with the target variable. These tests confirm whether observed patterns are statistically significant or could have occurred by chance.

7.1 Pearson Correlation — Numeric Features

Pearson correlation (r) measures the strength and direction of the linear relationship between each numeric feature and charges. Scipy's pearsonr() was used to also extract the p-value, which confirms statistical significance.

Threshold: $p < 0.05$ (95% confidence the relationship is real, not random noise).

Feature	Correlation (r)	P-Value	Interpretation
is_smoker	**0.787**	< 0.001	Strongest driver — highly significant
age	0.299	< 0.001	Moderate positive — significant
bmi	0.198	< 0.001	Weak positive — significant
bmi_category_Obese	0.155	< 0.001	Weak — significant
children	0.068	0.013	Very weak but technically significant
is_female	−0.057	0.037	Near-zero — borderline
region columns	< 0.05	> 0.05	Weak — may not contribute meaningfully

7.2 Chi-Square Test — Categorical Features

The Chi-Square Test of Independence (χ^2) was used to test whether categorical features are statistically linked to charges. Since charges is continuous, it was first binned into 4 quartile groups using pd.qcut(), creating an ordinal target for the test.

For each categorical feature, a contingency table (observed frequency grid) was compared against an expected table derived from pure random chance. A p-value < 0.05 means the feature is statistically dependent on the charge quartile — i.e., the feature carries predictive information.

Feature	χ^2 Statistic	P-Value	Decision
is_smoker	High	< 0.001	**Reject Null — Keep**
bmi_category_Obese	High	< 0.001	**Reject Null — Keep**
bmi_category_Overweight	Moderate	< 0.05	**Reject Null — Keep**
region_southeast	Moderate	< 0.05	**Reject Null — Keep**

is_female	Low	> 0.05	Accept Null — Weak signal
region_northwest / southwest	Low	> 0.05	Accept Null — Weak signal

Key insight from both tests: is_smoker is overwhelmingly the most important feature. BMI-related features (raw + category) rank second. Region has weak statistical significance and may be dropped or deprioritised in model selection.

8. Train-Test Split & Feature Scaling

8.1 Data Leakage — Why Split Comes First

A critical principle in machine learning preprocessing is that the test set must remain entirely unseen until final model evaluation. Any operation that uses statistics derived from the full dataset — including the test rows — before the split constitutes data leakage.

In this project, StandardScaler computes the mean (μ) and standard deviation (σ) of each column. If applied before splitting, the scaler incorporates test row values into those statistics. The model is then trained on data that has been influenced by the test set — inflating apparent performance metrics while degrading real-world reliability.

Correction applied: the original premature scaler call was commented out and moved to after the train-test split.

8.2 Train-Test Split

The dataset was split into a training set (80%) and a test set (20%) using sklearn's train_test_split with random_state=42 for reproducibility.

Set	Rows	Purpose
X_train / y_train	~1,070 rows (80%)	Model learns from these
X_test / y_test	~267 rows (20%)	Never seen during training — used only for final evaluation

Target variable used: **log_charges** (log-transformed version). Predictions are converted back to dollars using np.exp() during evaluation.

8.3 Feature Scaling — StandardScaler

StandardScaler was applied to the three continuous numeric columns: age, bmi, and children. Binary and one-hot encoded columns (0/1) do not require scaling.

$Z = (x - \mu) / \sigma$ — transforms each value to number of standard deviations from mean

- scaler.fit_transform(X_train): learns μ and σ from training rows only, then scales them.
- scaler.transform(X_test): applies the same μ and σ to test rows — does not re-learn. Test rows never influence the scaler's statistics.

This ensures the model operates on a level playing field across features. Without scaling, the age column (range 18–64) and charges column (range $1k–$64k) would have disproportionate influence on distance-based or gradient-based models purely due to magnitude differences.

9. Preprocessing Summary

The following table summarises all steps applied to the dataset from raw input to model-ready feature matrix:

#	Step	Action	Outcome
1	Missing Values	isnull().sum() check	None found
2	Duplicate Removal	drop_duplicates()	1 row removed → 1,337 rows
3	Label Verification	value_counts() on categoricals	All clean, no typos
4	Outlier Capping	IQR clip on bmi and charges	Extremes bounded, rows preserved
5	Label Encoding	sex → is_female, smoker → is_smoker	Binary text → 0/1 integers
6	One-Hot Encoding	get_dummies on region, drop_first=True	3 new region columns
7	BMI Binning	pd.cut() + get_dummies on bmi_category	3 new medical category columns
8	Log Transform	log_charges = np.log(charges)	Skewness 1.5 → ~0.2
9	Train-Test Split	train_test_split 80/20, random_state=42	1,070 train / 267 test rows
10	Feature Scaling	StandardScaler fit on train only	age, bmi, children standardised

9.1 Final Feature Matrix

After all preprocessing steps, the model-ready feature matrix X_train contains the following 11 columns:

Feature	Type	Notes
age	Scaled numeric	Z-score standardised
is_female	Binary (0/1)	Label encoded from sex
bmi	Scaled numeric	Z-score standardised, outliers capped
children	Scaled numeric	Z-score standardised
is_smoker	Binary (0/1)	Label encoded from smoker
region_northwest	Binary (0/1)	One-hot encoded, northeast is reference
region_southeast	Binary (0/1)	One-hot encoded
region_southwest	Binary (0/1)	One-hot encoded

bmi_category_Normal	Binary (0/1)	BMI binned, Underweight is reference
bmi_category_Overweight	Binary (0/1)	BMI binned
bmi_category_Obese	Binary (0/1)	BMI binned — statistically significant in χ^2 test

9.2 Key Findings

- Smoking status (is_smoker) is the single strongest predictor of insurance charges — confirmed by both Pearson correlation ($r = 0.787$) and Chi-square test.
- Age is the second most important numeric predictor with a moderate positive correlation.
- BMI as a raw feature has weak correlation, but the derived bmi_category_Obese showed significant statistical dependence on charges — demonstrating the value of feature engineering.
- Gender (is_female) and most region columns show weak or statistically insignificant relationships with charges — candidates for dropping during model selection.
- The target variable log_charges has skewness reduced from 1.5 to ~0.2 — suitable for linear regression assumptions.

9.3 Next Steps

- Baseline model: Linear Regression on X_train / y_train (log_charges).
- Evaluation: RMSE and R^2 on X_test / y_test, predictions converted back via np.exp().
- VIF check: Verify no multicollinearity among numeric features before finalising the feature set.
- Advanced models: Random Forest or XGBoost to capture non-linear interactions (e.g., smoker × bmi interaction on charges).
- Feature importance: Post-model analysis to confirm or remove low-signal features identified in the chi-square and Pearson tests.